

Resource Center and Technology Watch (Updated 2025)

This document lists the technical resources and tutorials consulted for the completion of the project. **RT Climatometer**. It demonstrates the learning and technological monitoring process undertaken.

Video Tutorials (2024-2025 Selection)

Backend : Flask & Python

- **[Tutorial] Flask Step-by-Step (Microsoft Learn 2025):** Complete guide on structuring modern Flask projects and managing virtual environments.
[Consult the interactive documentation \(https://learn.microsoft.com/fr-fr/visualstudio/python/learn-flask-visual-studio-step-01-project-solution\)](https://learn.microsoft.com/fr-fr/visualstudio/python/learn-flask-visual-studio-step-01-project-solution)
- **[Video] SQLite3 with Python (January 2023):** Tutorial on the interaction between Python and SQLite3, ideal for understanding CRUD and cursor management.
[Lien YouTube \(https://www.youtube.com/watch?v=JiEoZ8Z9oUQ\)](https://www.youtube.com/watch?v=JiEoZ8Z9oUQ)
- **[Video] Python Threading & Background Tasks:** Understanding how to run processes in parallel so as not to block the web server during API calls.
[Docstring technical explanation \(https://www.docstring.fr/glossaire/threading/\)](https://www.docstring.fr/glossaire/threading/)

Frontend: Leaflet & Mapping

- **[Video] JavaScript: Create a geographic map with Leaflet:** Practical tutorial to integrate a map, markers and manage zoom.
[Lien YouTube \(https://www.youtube.com/watch?v=xxuTFUWA9Nk\)](https://www.youtube.com/watch?v=xxuTFUWA9Nk)
 - **[Guide] Complete Leaflet - GIS & Territories Blog:** A structured course on conditional display and dynamic symbols on a web map.
[Read the tutorial \(https://www.sigterritoires.fr/index.php/tutoriel-leaflet/\)](https://www.sigterritoires.fr/index.php/tutoriel-leaflet/)
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Official Documentation & API

Weather API (OpenWeatherMap)

- **API Documentation 2024: Guide on key-based authentication (Limiting) to avoid HTTP 429 errors.** appid and request limit management (Rate)
[Documentation API OpenWeather \(https://openweathermap.org/api\)](https://openweathermap.org/api)
- **Security guide: Best practices for hiding API keys and using local caching (implemented in our project to limit calls).**
[YouTube : How to use API keys safely \(https://www.youtube.com/watch?v=HJvlgpoZNbQ\)](https://www.youtube.com/watch?v=HJvlgpoZNbQ)

Framework & SQL

- **Flask Quickstart (v2.1+): Reference for rendering Jinja2 templates and managing objects** request.
[Documentation Flask FR \(https://flask-fr.readthedocs.io/quickstart/\)](https://flask-fr.readthedocs.io/quickstart/)
- **SQLite Python Docs: Official reference on the use of the SQL** sqlite3 and security against injections module.
[Python.org - sqlite3 module \(https://docs.python.org/3/library/sqlite3.html\)](https://docs.python.org/3/library/sqlite3.html)

Development Tools

- **VS Code (Microsoft)** :Main IDE with integrated Git manager.
- **DB Browser for SQLite** : Tool used to view tables residences and real-time weather readings.
- **Postman** Used to test the JSON responses from the OpenWeather API before integration into the script
update_weather.py

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